

IoT-Zoo: A Container-Based Framework for Heterogeneous IoT Device Profiles and Reproducible Traffic Capture

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<https://gt-iotedu.github.io/>

Contextualization

- IoT research depends on **realistic and reproducible experimental data**.
- Real deployments combine many **domains, devices, payloads,** and **communication** patterns.
- Security and networking solutions need traffic that **reflects this heterogeneity**.

Current Gap

- Existing testbeds often **prioritize scale over diversity**.
- Many platforms scale by **replicating a small number** of device types.
- This limits the representation of real IoT behavior, such as:
 - sensing modalities;
 - update rhythms;
 - payload structures; and
 - protocol stacks.

IoT testbeds and traffic generation frameworks

Testbed / Project	Diversity
TON_IoT	10 sensors/devices
Gotham	11 distinct profiles
GothX	11 distinct profiles
DoS/DDoS MQTT	13 sensor types

IoT-Zoo: 43 distinct profiles

(e.g., environmental, smart building, industrial, smart agriculture, human-centric, cameras)

Proposal: IoT-Zoo

43



3.3×



+231%

distinct profiles
in IoT-Zoo

higher diversity than the
second most diverse
compared platform

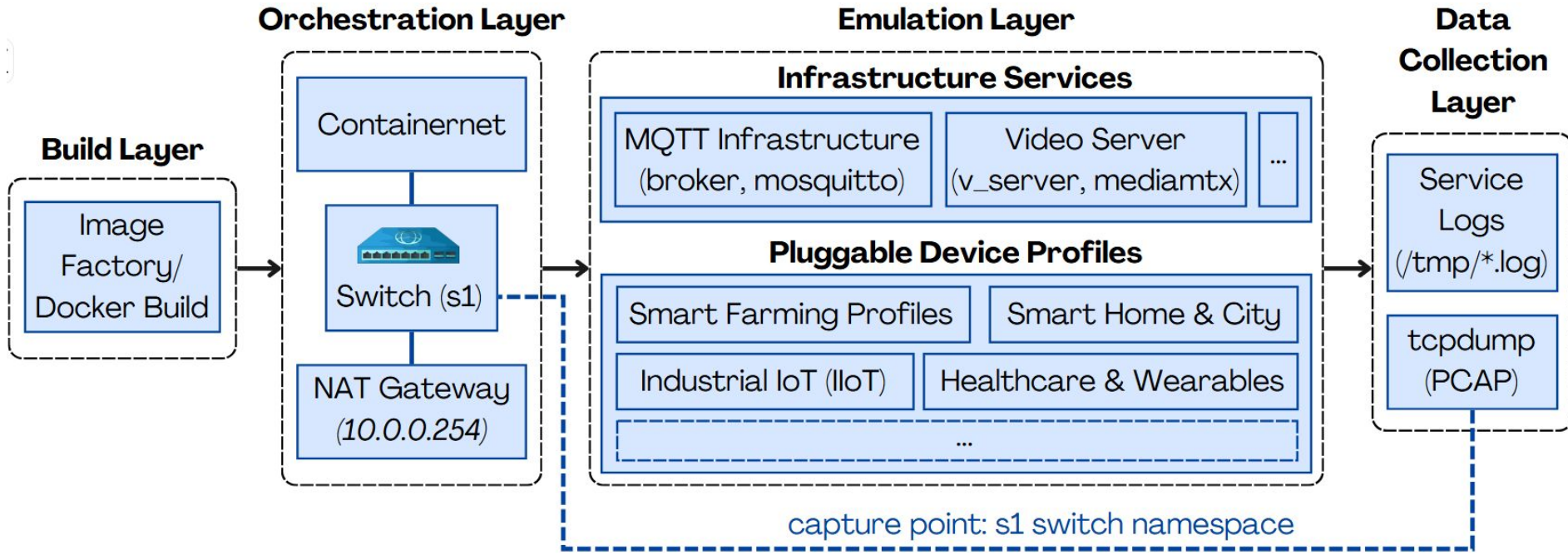
increase in
profile diversity

What does IoT-Zoo provide?

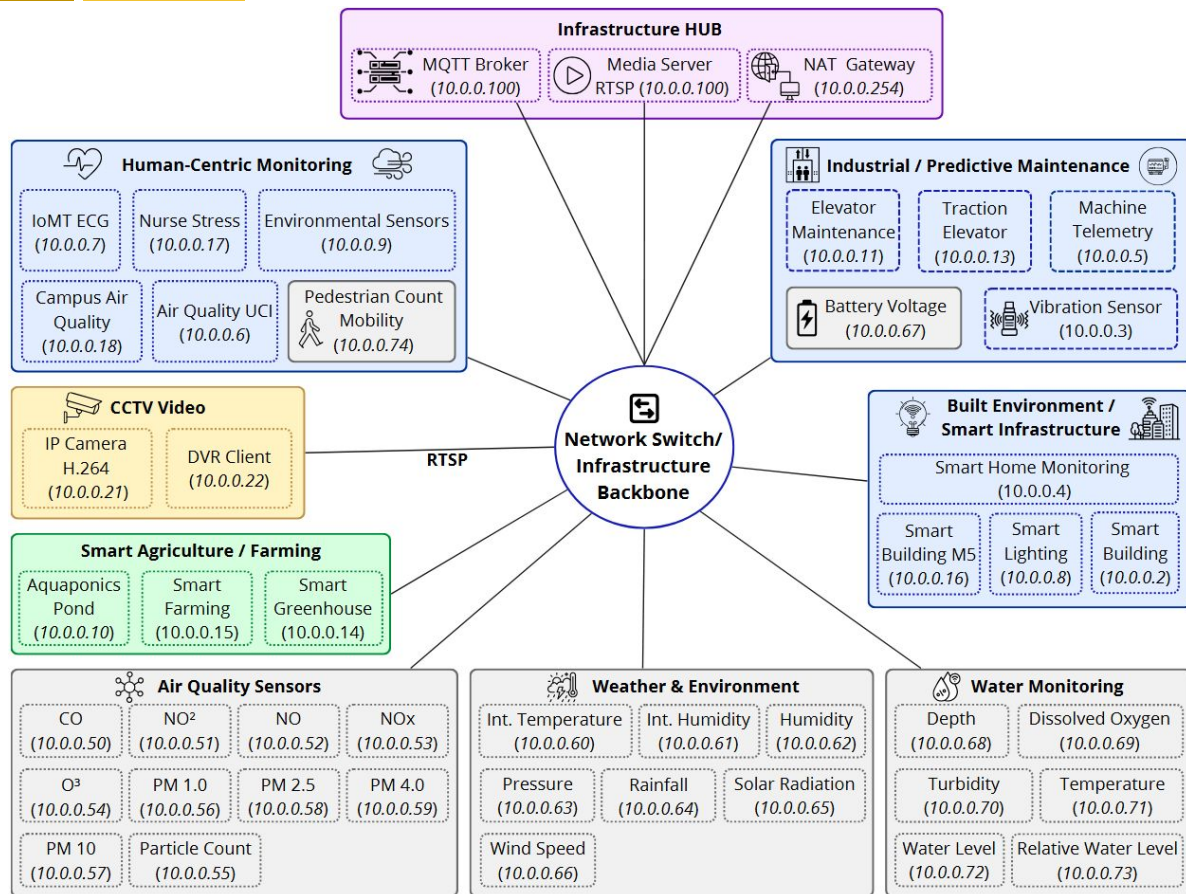
Proposal: IoT-Zoo

- **Dataset-driven IoT profiles** encapsulated as **Docker** containers.
- **Automated orchestration** with Containernet and a virtual network topology.
- Real application protocols, including **MQTT telemetry** and **RTSP video streams**.
- Single-command execution with **automated PCAP capture** and **service logs**.

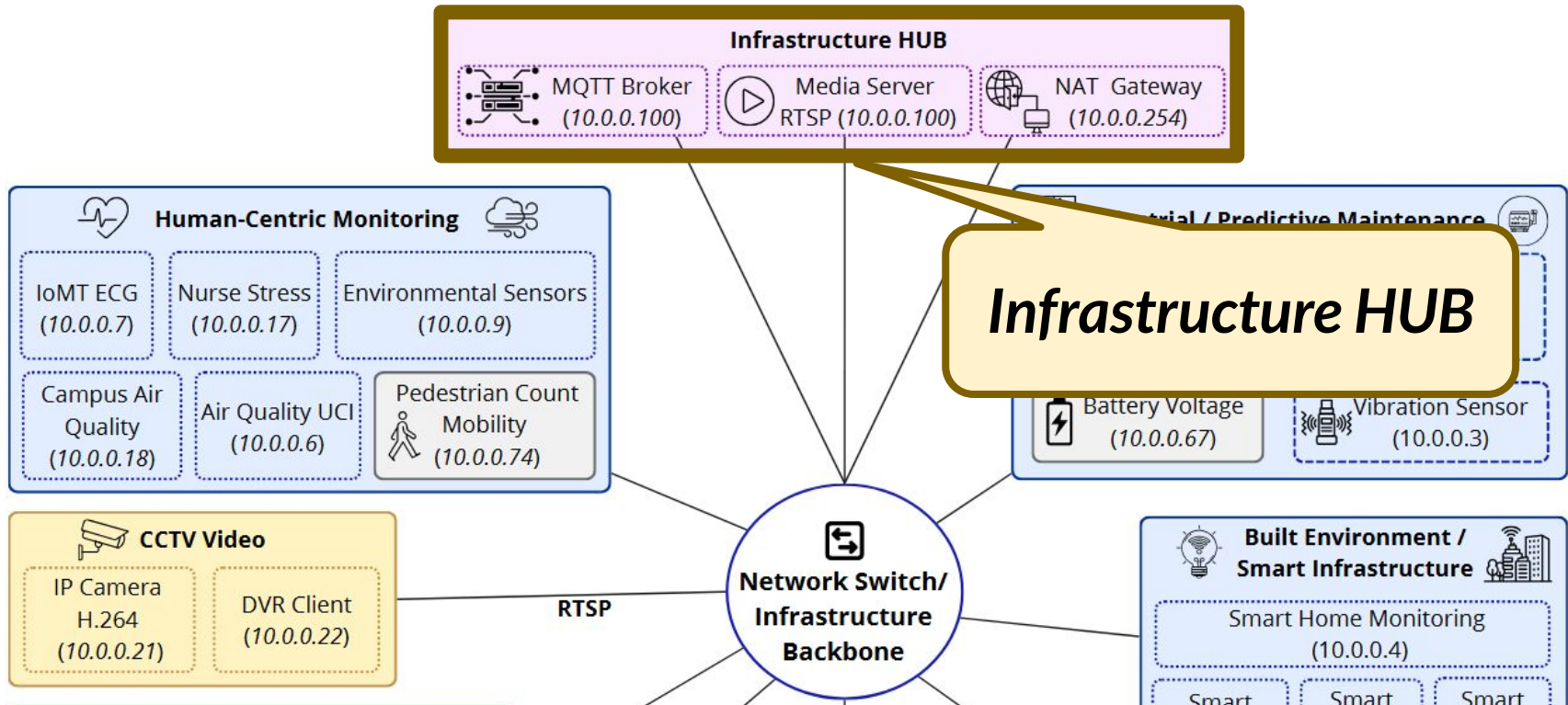
Proposed Architecture



Reference Implementation of the IoT-Zoo

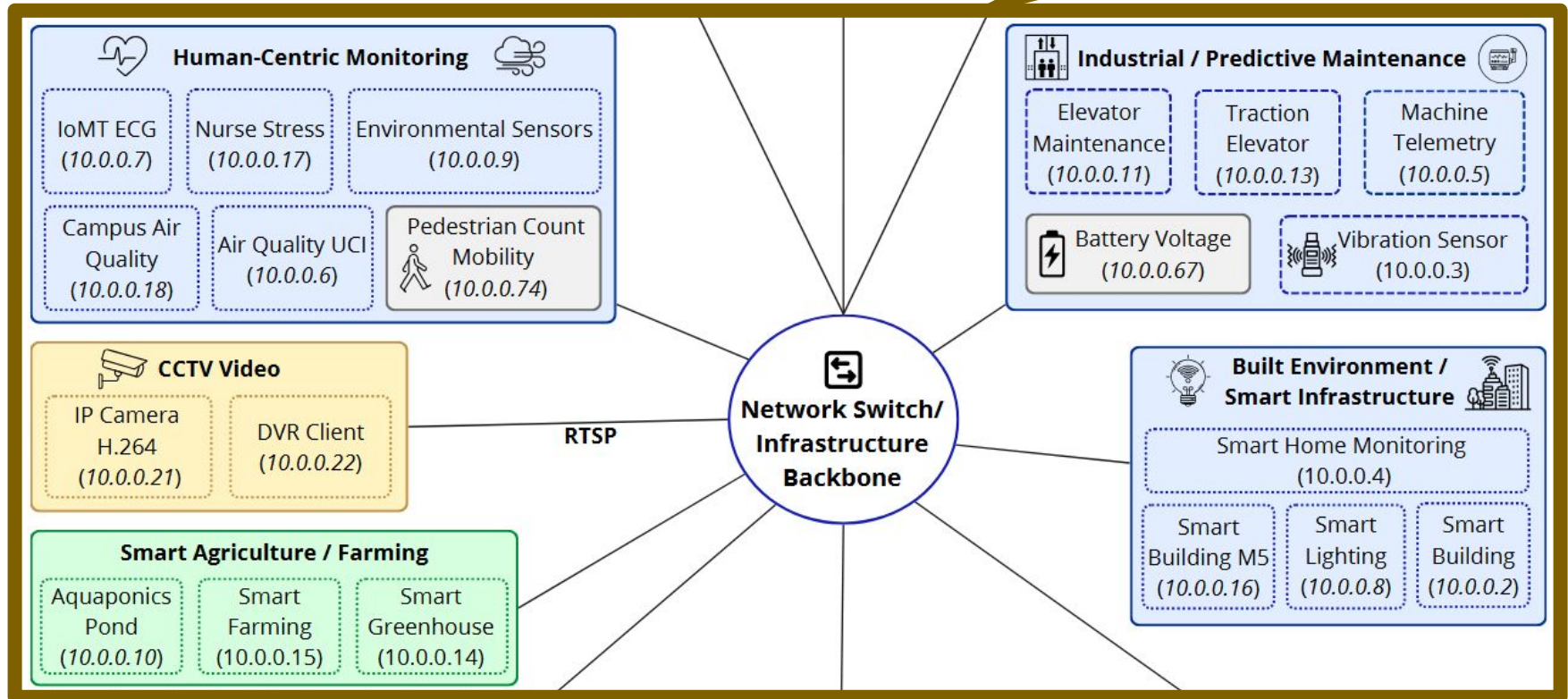


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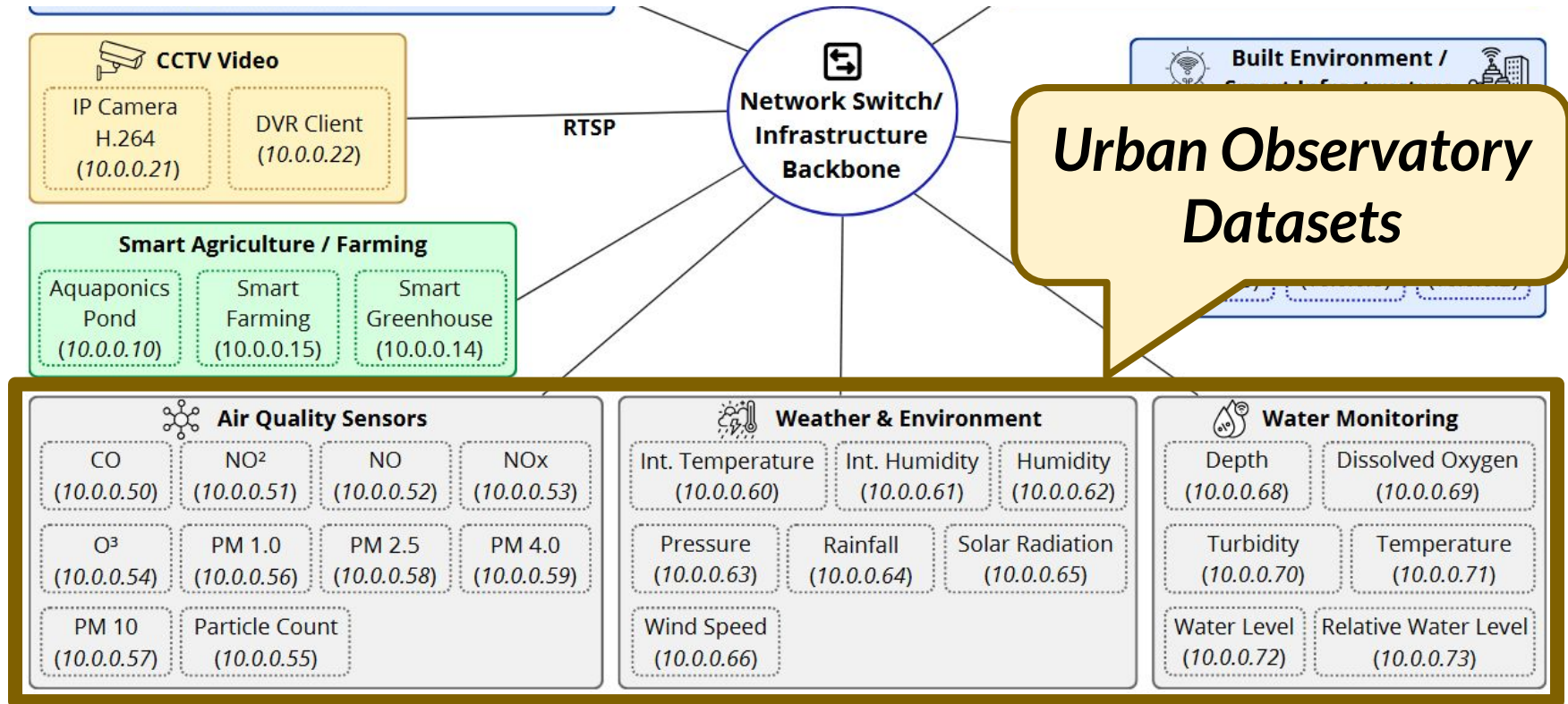


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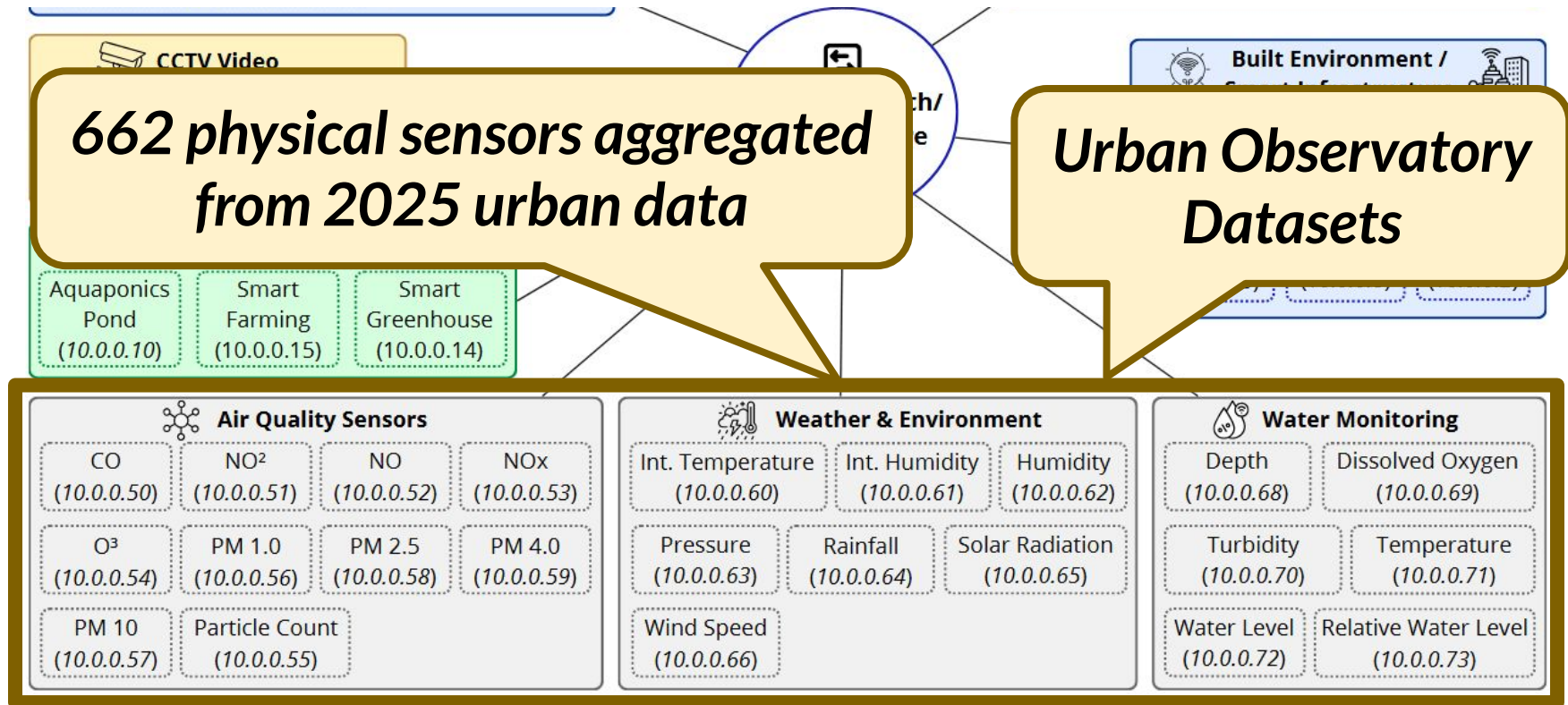
Public IoT Datasets



Reference Implementation of the IoT-Zoo



Reference Implementation of the IoT-Zoo



Network metrics for IoT device profiles

Device	Pkts	IAT(s)	Jit(s)	Device	Pkts	IAT(s)	Jit(s)
Building Rooms	269	2.2111	2.5412	Urban Air: PM10	2989	0.2003	0.3415
Build: Int. Hum.	301	1.9987	1.6460	Urban Air: PM2.5	3055	0.1960	0.3390
Build: Int. Temp.	302	1.9857	1.6513	Urban Air: PM4	2462	0.2433	0.4289
Ind. Motor (Cooler)	121	4.9588	0.4547	Urban Air: Particle	2450	0.2445	0.4297
Legacy Air Sensor	74	7.9919	6.2948	Water L: Abs	1812	0.3292	1.2396
Mobility Sensor	2292	0.2605	1.8041	Water L: Rel	1380	0.4322	1.4048
Predictive Maint.	120	4.9999	0.0993	Water Q: DO	180	3.3274	2.3576
Smart Home (Domotic)	78	7.6615	4.9212	Water Q: Depth	238	2.5134	2.5001
Smart Lighting	125	4.7996	0.9390	Water Q: Temp	208	2.8774	2.4701
Urban Air: CO	2338	0.2562	0.4364	Water Q: Turb	308	1.9403	2.4356
Urban Air: NO	762	0.7829	1.8172	Weather: Hum	302	1.9920	1.6473
Urban Air: NO2	3519	0.1702	0.3228	Weather: Press	3843	0.1560	0.3623
Urban Air: NOx	597	0.9996	2.0002	Weather: Rain	1352	0.4411	1.4181
Urban Air: O3	2894	0.2070	0.4050	Weather: Solar	592	1.0137	0.9991
Urban Air: PM1	2999	0.1997	0.3411	Weather: Wind	1010	0.5940	0.8909
e-Health ECG	1184	0.5037	0.3834	Human: Nurse Stress	124	4.8572	1.0376
Aquaponics	117	5.1495	0.9955	Smart Building M5	123	4.8955	0.9522
Environmental Sensors	121	4.9954	0.9436	Farming Sensor	120	4.9711	0.9409
Greenhouse	237	2.5146	1.3538	Ind. Pred. Maint. GW	119	5.0256	0.9107
				Ind. Traction Elevator	122	4.9423	0.9407

Periodic telemetry | High-volume streams | Bursty behavior

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Bursty behavior

Resource Consumption Profile by Device Category

Device Category	Total Profiles	Mem. Usage (MiB) ($\mu \pm \sigma$)	CPU Peak (%)
Air Quality Sensors	10	48.65 ± 1.98	11.75
Weather & Environment	5	48.65 ± 2.02	12.04
Water Monitoring	6	48.62 ± 2.03	11.26
Built Environment / Smart Infra.	8	20.82 ± 16.24	38.41
Human-Centric Monitoring	3	27.18 ± 16.37	7.71
Industrial / Predictive Maintenance	6	16.83 ± 12.15	7.67
Smart Agriculture / Farming	3	14.58 ± 4.43	9.93
CCTV Video	2	8.94 ± 6.53	5.27
Infrastructure HUB	3	12.44 ± 10.01	1.60
Total / Average	46	≈ 27.42 MiB	-

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Low memory footprint

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Controlled CPU usage

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Lightweight and predictable resource footprint with 46 heterogeneous containerized profiles on commodity hardware: 4 vCPUs and 8 GB RAM.

Final Remarks

- IoT-Zoo is a **container-based testbed** for reproducible IoT experimentation.
- It supports **43 heterogeneous IoT device profiles** across multiple domains.
- The framework automates **topology deployment, traffic generation, and PCAP capture**.
- Results indicate that IoT-Zoo can reproduce diverse traffic patterns with **low resource consumption**.

Future Work

- Make the topology **parametrizable**, allowing users to define the **number of nodes**, **gateways**, and **scenario** configurations.
- Integrate **new communication protocols**, such as CoAP, Zenoh, and XRCE-DDS.
- Add **more IoT device profiles** from different application domains.
- Include **attack scenarios** with ground-truth labels for security and intrusion detection experiments.

Thank you! Questions?



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